Modeling of Electrical Systems

(电路系统的建模)

Main Contents of Course (课程主要内容)

Mechanical (机械)

Electrical (电路)

Electro-Mechanical (机–电)

Fluid (流体)

Thermal (热系统)

Dynamic Modeling (动态建模)

Model structure (模型结构)

System Identification(系统辨识)

Parameter estimation (参数估计)

Control System Design(控制系统设计)

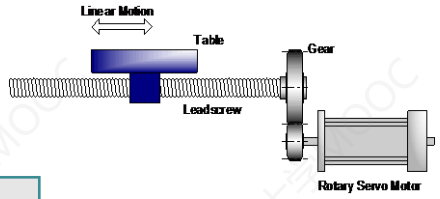
System Analysis (系统分析)

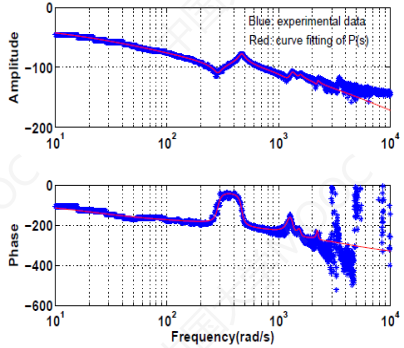
Root-locus design (根轨迹设计)

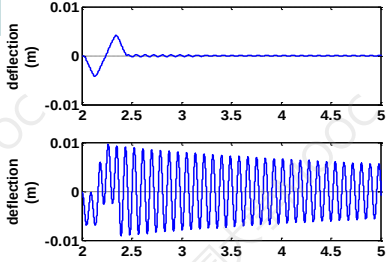
Pole placement design (极点配置)

Design limitations (性能局限)

Architectural issues (控制结构)







Variables

- ***q*** : charge (电荷)[C] (Couloimb)
- ***i*** : current (电流) [A]
- ***e*** : voltage (电压)[V]

$$\frac{d}{dt}q = i$$

$$q(t_1) = q(t_0) + \int_{t_0}^{t_1} i(t)dt$$

- ***R***: resistance (电阻) [Ω]
- ***C***: capacitance (电容) [Farad]
- ***L***: inductance (电感) [H] (Henry)

- ***p*** : power (功率) [Watt]
- ***w***: work (功) (energy (能量)) [J]

1 [J] (Joule) = 1 [V.A.sec]

$$p = e \cdot i$$

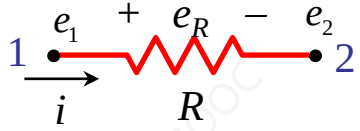
$$w(t_1) = w(t_0) + \int_{t_0}^{t_1} p(t)dt$$

$$= w(t_0) + \int_{t_0}^{t_1} (e.i)dt$$

Basic Modeling Elements

- **Resistor (电阻)**
  - Ohms Law (欧姆定律)
    Voltage across is proportional to the through current (电压正比于流过的电流)

$$e_{12} = e_1 - e_2 = e_R = R\,i \quad \Leftrightarrow \quad i = \frac{1}{R}\,e_R$$



Static relation (静态关系)

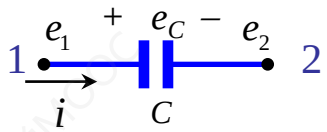
  - Dissipates energy through heat(通过热消耗能量)

$$p = R\,i^2 = \frac{1}{R}\,e^2$$

  - Analogous to friction elements (B=R) in mechanical systems, e.g. dampers (类似于机械系统的摩擦元件，如阻尼器)

- **Capacitor (电容)**
  - Charge collected is proportional to the voltage across (电荷正比于电压)
  - Current is proportional to the rate of change of the voltage across (电流正比于电压变化率)

$$q = C\,e_C \quad \Leftrightarrow \quad i = C\left(\frac{d}{dt}e_C\right) = C\left(\frac{d}{dt}e_{12}\right)$$



dynamic relation (动态关系)

  - Energy supplied is stored in its electric field and can affect future circuit response (电场中存贮的能量会影响电流响应)
  - Analogous to spring elements (C=1/k) in mechanical systems (类似于机械系统的弹簧元件)

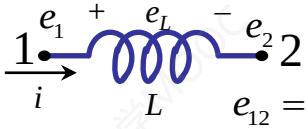
Analogous: $q = x, i = \dot{x}, e_c = F$

Basic Modeling Elements

- **Inductor** (电感)
  - Voltage across is proportional to the rate of the change of the through current (通过的电压正比于流过电流的变化率).

$$e_{12} = e_L = L\left(\frac{d}{dt}i\right)$$

Dynamic (动态关系)



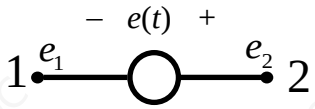
$$e_{12} = e_1 - e_2$$

  - Energy supplied is stored in its magnetic field (所提供能量存储在磁场中).

$$w = \frac{1}{2} L\, i^2$$

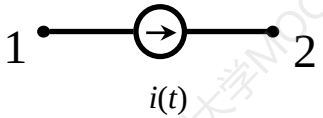
  - Analogous to mass elements (L=M) in mechanical systems (类比于机械系统的质量元件).

- **Voltage Source** (电压源)
  - Maintain specified voltage across two points, regardless of the required current (不管电流变化能维持两端点电压不变)



$$e_{12} = e_1 - e_2 = -e(t)$$

- **Current Source** (电流源)
  - Maintain specified current, regardless of the required voltage (不管电压变化能维持流过的电流不变)

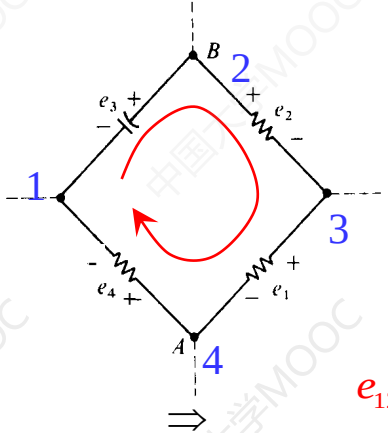


Interconnection Laws

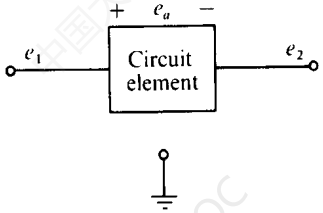
- **Kirchhoff's Voltage Law (loop law)**
  - The total voltage drop along any *closed loop* in the circuit is zero

  (基尔霍夫电压定律：回路闭环压降为零).

$$\sum_{\text{Closed Loop}} e_j = 0$$



$$e_a = e_{12} = e_1 - e_2$$



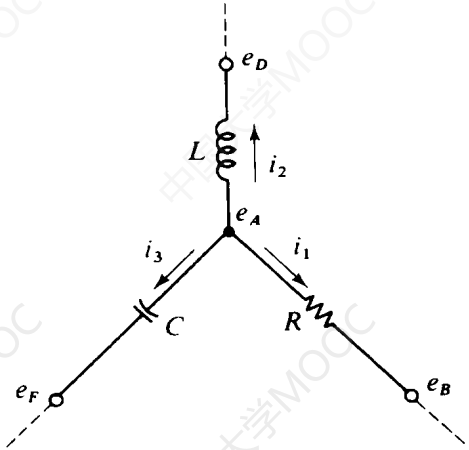
$$e_{12} + e_{23} + e_{34} + e_{41} = 0$$

$=e_i$

- **Kirchhoff's Current Law (node law)**
  - The algebraic sum of the currents at any node in the circuit is zero

  (基尔霍夫电流定律：节点电流代数和为零).

$$\sum_{\text{Any Node}} i_j = 0$$



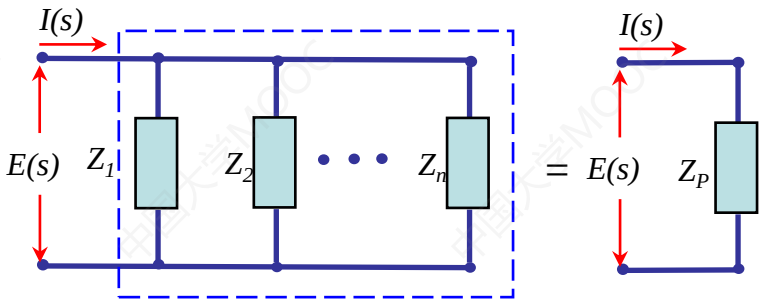
$$i_1 + i_2 + i_3 = 0$$

Series and Parallel Elements

- **Parallel combinations** (并联)

Same Voltage across elements

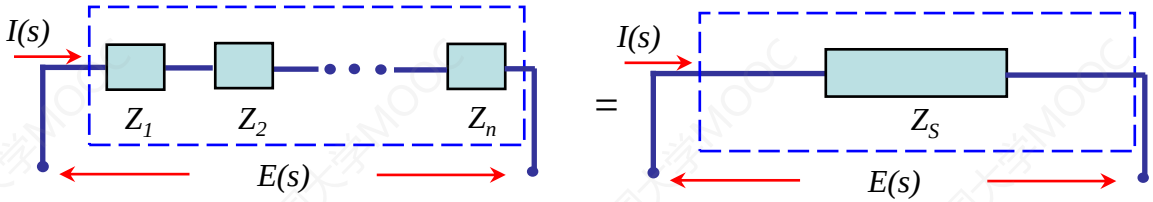
$$\Delta(Voltage_i) = \Delta(Voltage_j)$$



- **Series combinations** (串联)

Same Current through elements

$$(Current_i) = (Current_j)$$



- **Equivalent Complex Impedance** (等效阻抗)

$$\frac{1}{Z_P(s)} = \frac{1}{Z_1(s)} + \frac{1}{Z_2(s)} + \cdots + \frac{1}{Z_n(s)}$$

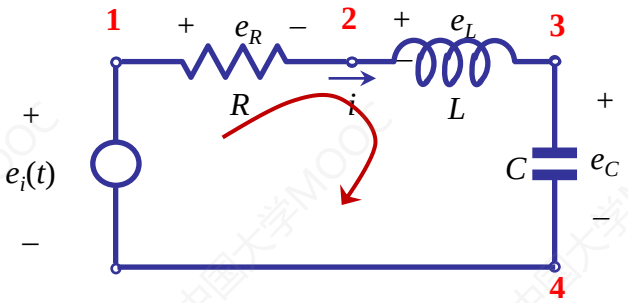
$$Z_s(s) = Z_1(s) + Z_2(s) + \cdots + Z_n(s)$$

Modeling Steps

- **Understand System Function and Identify Input/Output Variables (理解系统功能和定义输入/输出量)**
- **Draw Simplified Schematics Using Basic Elements (画出由基本元件组成的系统简化原理图)**
- **Develop Mathematical Model (建立数学模型)**
  - Label Each Element and the Corresponding Voltages (标出每个元件和相应的电压).
  - Label Each Node and the Corresponding Currents (标出每个节点和相应的电流)
  - Apply Interconnection Laws (应用物理规律和互联关系)
  - Check that the Number of Unknown Variables equals the Number of Equations (检查未知变量个数和方程个数是否相等)
  - Eliminate Intermediate Variables to Obtain Standard Forms (消除中间变量来获得标准形式的模型方程)

In Class Exercise

Ex1: Derive the I/O model for the following circuit. Let voltage $e_i(t)$ be the input and the voltage across the capacitor be the output **(推导下述电路的I/O模型，$e_i(t)$ 是输入，电容上的电压为输出)**.



Element Laws:

$$e_{12} = R.i$$

$$e_{23} = L\frac{di}{dt}$$

$$i = C\frac{de_{34}}{dt}$$

➢**Kirchhoff's Loop Law:**

$$e_{12} + e_{23} + e_{34} + \underbrace{e_{41}}_{=-e_i} = 0$$

➢**No. of Unknowns:**

$$e_{12},\ i,\ e_{23},\ e_{34} = y$$

➢**Simplify (eliminate e_{12},e_{23},i):**

$$e_{12}, e_{23}: \quad \Rightarrow R.i + L\frac{di}{dt} + e_{34} - e_i = 0$$

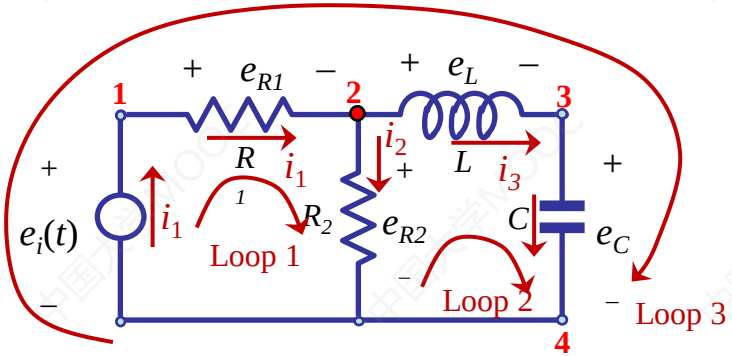
$$i: \quad \Rightarrow RC\frac{de_{34}}{dt} + LC\frac{d^2e_{34}}{dt^2} + e_{34} = e_i$$

➢**I/O Model:**

$$LC\ddot{e}_{34} + RC\dot{e}_{34} + e_{34} = e_i$$

Example

Ex: Obtain the I/O model for the following circuit. The input is the voltage $e_i(t)$ of the voltage source and the through current of the inductor is the output **(推导下列电路的I/O模型，电压源$e_i(t)$是输入，电感上电流是输出)**.



➢Elemental Equations:

$$e_{12} = R_1 i_1 \qquad (1)$$

$$e_{24} = R_2 i_2 \qquad (2)$$

$$e_{23} = L\frac{di_3}{dt} \qquad (3)$$

$$i_3 = C\frac{de_{34}}{dt} \qquad (4)$$

➢Voltage Law

Loop 1: $e_{12} + e_{24} + e_{41} = 0 \qquad (5)$

Loop 2: $e_{23} + e_{34} + \overset{-e_i}{(-e_{24})} = 0 \qquad (6)$

➢Current Law

Nodal: $i_1 - i_2 - i_3 = 0 \qquad (7)$

➢Unknown Variables

$e_{12}, i_1, e_{24}, i_2, e_{23}, i_3 = y, e_{34}$, 7 UVs = 7 eqs,

Example (cont.)

➢**Eliminating e_{12},e_{24},e_{23} (substituting eqs (1),(2),(3) into the rest equations)**

$$\left.\begin{aligned}
&(4):\quad i_3 = C\frac{de_{34}}{dt}\\
&(5):\quad i_1R_1 + i_2R_2 - e_i = 0\\
&(6):\quad L\frac{di_3}{dt} + e_{34} - i_2R_2 = 0\\
&(7):\quad i_1 - i_2 - i_3 = 0
\end{aligned}\right\}$$

e_{34}, i_1 (substituting eqs(6),(7) into (4),(5))

$$\begin{cases} e_{34} = i_2R_2 - L\dfrac{di_3}{dt} \\ i_1 = i_2 + i_3 \end{cases}$$

$$\Rightarrow \begin{cases} i_3 = CR_2\dfrac{di_2}{dt} - CL\dfrac{d^2i_3}{dt^2} & (4) \\ i_2R_1 + i_3R_1 + i_2R_2 = e_i & (5) \end{cases}$$

$$\Rightarrow \quad \text{from(5):}\quad i_2 = \frac{1}{(R_1+R_2)}\left[e_i - i_3R_1\right]$$

$$\text{into(4):}\quad i_3 = CR_2\frac{1}{R_1+R_2}\left[\dot{e}_i - R_1\frac{di_3}{dt}\right] - CL\frac{d^2i_3}{dt^2}$$

➢**I/O Model**

$$LC\frac{d^2i_3}{dt^2} + \frac{CR_1R_2}{R_1+R_2}\frac{di_3}{dt} + i_3 = \frac{R_2C}{R_1+R_2}\dot{e}_i$$

Summary

- **Basic Modeling Elements (基本建模元件)**
- **Interconnection Relationships (互联关系)**
- **Derive Input/Output Models (推导输入/输出模型)**